

The Nature of UX Design

Highlights

- What is design?
- Switch in mode of thinking from understanding needs (UX research) to design
- Universality of design and relationships to other fields
- Ambiguity of scope
- The purpose of design: To satisfy human needs
- Pyramid of human needs
- Stages of design
- The overall design-creation lifecycle to illustrate expanding granularity of substance and richness of detail through iterations of:
 - Generative design
 - Intermediate design
 - Detailed design
 - Design refinement
- Design lifecycle for the agile UX funnel

11.1 INTRODUCTION

11.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 11.1). In this chapter, we introduce the UX design solutions lifecycle activity with a discussion of the nature of design.

This chapter contains a descriptive discussion of the characteristics of design, especially UX design. The how-to part of design is in subsequent chapters.

11.1.2 What Is Design?

The topics of design, what it is, and how to do it have all been studied in many different domains over time. Design is perhaps the original embodied skill of humans, starting with the fashioning of artifacts in one’s environment into tools.

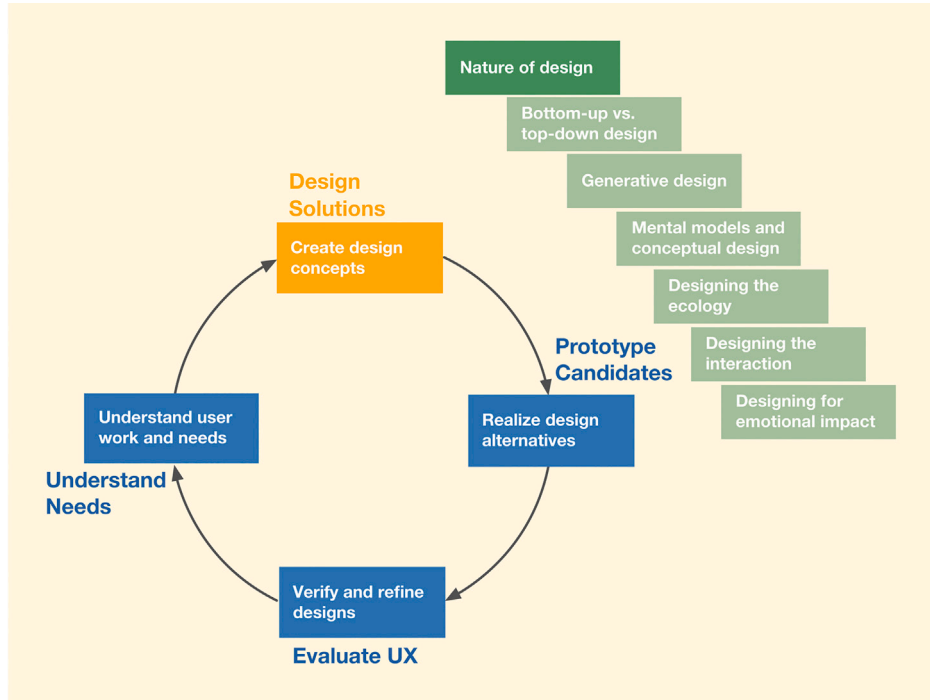


Fig. 11.1

You are here in the nature of design chapter within the design solutions activity, in the context of the overall Wheel UX lifecycle.

The study of design has retained its relevance through time, manifested in a wide variety of perspectives, practices, and trends. A graphic or visual designer on the design team may think about design in terms of emotion, joy, and art. A usability analyst may think about design from a diagnostic perspective. Consultants and design agencies may look at design from the perspective of what sells, who is going to pay for it, and how to budget for it. Another common view is that design is about form (shape of the product) and function (purpose of the product). These perspectives aside, in this book we focus on two ways of thinking about design.

11.1.2.1 Two ways the word design is used

Most English dictionaries define the word *design* along two main dimensions: as a verb (the act of creating) and as a noun (the resulting concept of, or plan for, a product or system). For example, the “pinch to zoom” on an iPhone is an element (design as a noun) that was created as the result of a process of creation (design as a verb). Fig. 11.2 shows the simple relationship between these ways of using design.

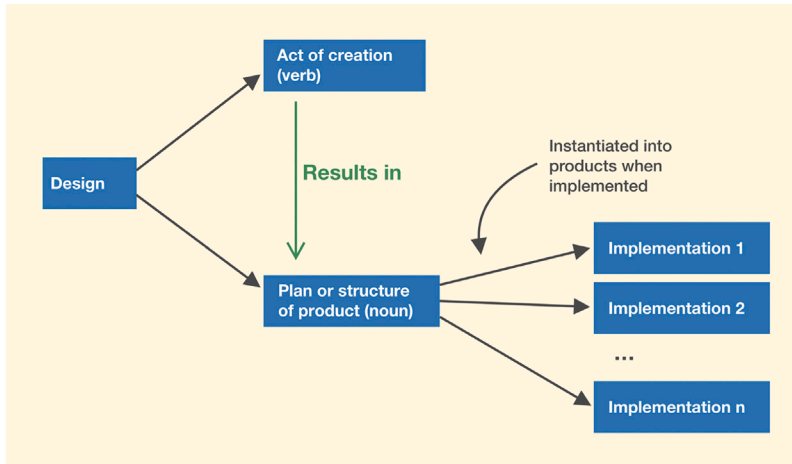


Fig. 11.2
Design as a noun and
design as a verb.

11.1.2.2 Design as a verb

Design as a verb refers to the act of creating new designs and design ideas that did not exist before—solutions to known problems and solutions looking for problems. This is the design solutions box in the overall lifecycle (see Fig. 11.1), within which designs are created.

11.1.2.3 Design as a noun

A definition of design as a noun is the concept or plan for a product or system. It is about the organization, composition, or structure of elements to be executed or constructed. When we hear someone say, “I like this design” or “This design is confusing,” the person is referring to the design concept behind the product. Unpacking this definition further, design as a noun is specifically about an abstract construct that represents the way a product is conceived by its designer, a description of the plan or structure of the product.

It is not about an instantiation of that construct in the form of a tangible object or system, which can be thought of as one possible implementation of the design. In fact, not all designs come to fruition when instantiation is attempted. Some plans are flawed, or are revealed as such, when put into action (either by simulating the result using a prototype or by constructing a real implementation). Unexpected constraints, unforeseen technical limitations, or other omissions yet to be discovered may force revisions to these plans.

11.1.3 Design Synthesis: Moving Across the Gap From Analysis to Design

In this part of the book, we switch our focus from UX research to design. This involves changing our perspective from the existing work¹ domain and work practice to an envisioned design domain and work practice. The transition to design is often regarded as the most difficult step in the UX lifecycle process, and we should expect it to be difficult. Beyer and Holtzblatt (1997) remind us “The design isn’t explicit in the data. The data guides, constrains, and suggests directions” that design “can respond to” (p. 218).

In other words, design is not simply a translation exercise; it is not about taking what we have learned about the existing work practice and recreating it using technology artifacts. There needs to be a switch in the mindset to focus on creating new solutions.

Wood (1998) calls the transition from analysis to design a step where magic happens. Kolko (2011) urges us to bring this magic to light. When we need this kind of magic to get us across a gap to the innovation of new design ideas, that is when we call on synthesis from the UX technique toolbox.

In [Section 7.3.1](#), we introduced the concept of synthesis as a means for making sense out of UX data and its analysis. There, synthesis meant pulling together multiple inputs from multiple sources and creating new ways to understand UX design needs and requirements that did not exist before in the raw UX research data. Another real payoff of synthesis occurs here, as a bridge from UX needs and requirements to UX design. This is still synthesis, but synthesis with a different goal. Here in design, synthesis is essential to creativity and innovation. It is where we combine things we know to spark the creation of new designs and design ideas.

11.1.4 Universality of Design and Relationship to Other Fields

Design is universal; it is about creating things (products, services, experiences) with the goal of helping humans satisfy a variety of needs.

Design is the core activity in many other creative fields. Fashion is about the design of clothes, which at a functional level is about protecting humans from the elements. But it is about much more. Clothing designers use colors, fabrics, and shapes to create fashion experiences in infinite variations.

Of all the design fields, perhaps industrial design is the one closest to what we do in UX. In the last two decades, with the proliferation of affordable

¹The term work broadly signifies anything people/users do while using products or systems (see [Section 6.2](#)).

technology in the form of handheld and wearable devices, industrial design and UX have become inextricably bound. The user experience of a smartphone or smartwatch is derived not just from the software user interface but also the shape, texture, materials, and form factor of the product that houses the user interface.

No matter the field, all these design endeavors share the same fundamental process activities of understanding, creating, prototyping, and evaluating. Each discipline's vocabulary and domain constructs may be different, but they are all ultimately about creating products to solve problems and satisfy human needs.

11.1.5 Design Thinking

The influence of design has even spread to business processes. Design thinking (Brown, 2008; Kolko, 2015a) is a business mindset of getting organizations to think like designers, applying design principles and practice to businesses and business processes. Design is more than just a box in a lifecycle; it is a separate discipline on its own. Design thinking is seeing everything in terms of design. Designers are called upon to create a new vision, seeking a profound and satisfying user experience (Knemeyer, 2015). Design thinking is market oriented. When I got my new iPhone, the label did not claim, “Made by Apple”; it claimed, “Designed by Apple”!

11.1.6 Relationship to Design in Architecture

Architecture is a field in which design is foremost, a field that stands as an inspiration for UX. Architecture is a discipline dedicated to designing spaces to support humans and their needs, to sustain—and even glorify—living and working. Spaces in this context include everything from cities, neighborhoods, houses, community places, and offices to the infrastructure that connects them.

Great designs in architecture do not just provide shelter; they have the potential to spawn vibrant communities and induce strong emotional responses. For example, when people walk on the sixth floor of Bloomberg's Global Headquarters in New York City (where Pardha used to work) and see the scene shown in [Fig. 11.3](#), they encounter a feeling of movement and awe.

Visitors to the Bloomberg building say they experience a feeling of high energy and fast pace. This is because every element in this space was carefully designed by architects to evoke that experience of dynamism and connectedness. The constant movement of people through the space and of digital displays showing the latest financial indicators, weather, and breaking news creates a sense of energy as you walk into this central atrium. The



Fig. 11.3
Beautiful space in the
Bloomberg building designed
to evoke dynamic energy.

expansive space and high ceilings combined with the curved glass surrounding an imposing courtyard inject a sense of grandness.

11.1.7 The Interdisciplinary Nature of Design

In [Chapter 1](#), we said that user experience is the sum total of all the effects felt by the user from what the user sees, does, hears, and feels and all the behaviors of the artifact during contact and communication between them. Given this breadth, it is expected that UX design teams have a wide variety of skills and backgrounds, including:

- Expertise in problem solving, analysis, and reasoning
- Expertise in constraint solving and optimization
- Expertise in product development, including estimation, budgeting, and timelines
- Subject-matter expertise in work domains and design platforms
- Design expertise in particular technologies
- Expertise in art, culture, liberal arts, and social sciences

11.1.8 Ambiguity in the Scope

A point of potential confusion arises from ambiguity in the scope of the process referred to by design. On one hand, we say that the UX design lifecycle embraces the entire scope of the UX design process of [Fig. 11.1](#),

calling it the UX design lifecycle process. So, one might conclude that this whole domain is the answer to the question: What is UX design?

The astute reader, however, may note there is also a box at the top of [Fig. 11.1](#) called design solutions within this lifecycle. Maybe that defines UX design. In fact, both interpretations of the term are useful, but we lack the vocabulary to distinguish them. We sidestep the ambiguity trap by following the loose conventions of the field and using design with both meanings, hoping that context will provide clarity.

Broad interpretation. In the broad interpretation, design refers to the overall UX design lifecycle process. In simple terms, the whole UX process is a design process, shown in blue toward the bottom of [Fig. 11.4](#).

Narrow interpretation. In the narrow interpretation, design refers to a single activity within the UX lifecycle, as shown in green toward the top of [Fig. 11.4](#). This narrower view also allows us to keep design separate from discussion of other lifecycle activities, such as understand needs or evaluate UX. As a lifecycle activity on its own, design is important enough to have its own definitions, activities, theory, and practice.

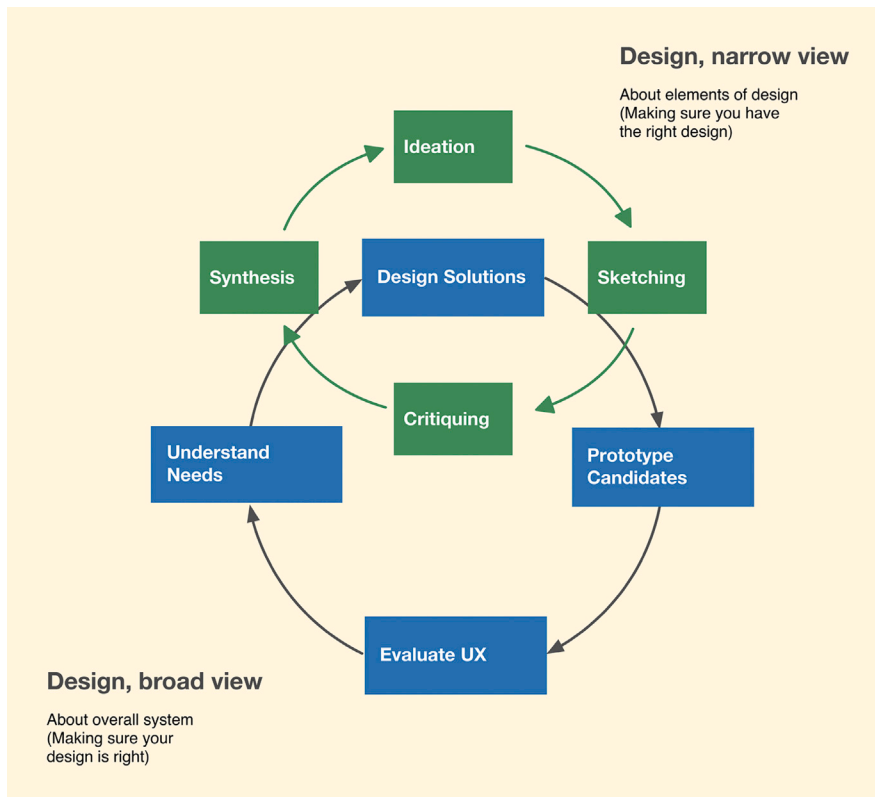


Fig. 11.4
 Comparing the narrow and broad views of design.

The narrow view has led to misconceptions. The narrow view, considered alone, may have contributed to a long-held misunderstanding of the role of UX designers in projects. As Brown (2008) says, “Historically, design has been treated as a downstream step in the development process—the point where designers, who have played no earlier role in the substantive work of innovation, come along and put a beautiful wrapper around the idea. Not seeing design, especially UX design, in the broad view, project teams have been slow to invite designers into the whole process” (p. 84).

Different views of design have led to different views of prototyping. The different views at different levels of UX design offer a good way to distinguish the two kinds of prototyping and evaluation that Buxton (2007b) points out:

1. *The narrow view of design.* The top (green) sub-lifecycle within the design-creation sub-lifecycle activity (see Fig. 11.4) is for the creative process of generating the right design (see Chapter 13). Ideation is about design creation and genesis, sketching creates multiple rapid low-fidelity prototypes, and critiquing is evaluation.
2. *The broad view of design.* Getting that design right is the goal of prototyping, evaluation, and iterative refinement within the broader overall UX lifecycle process, the bottom (blue) portion of Fig. 11.4. This larger lifecycle is where you expand the scope and fidelity of the designs created in the narrow view.

11.2 THE PURPOSE OF DESIGN: SATISFYING HUMAN NEEDS

11.2.1 A Pyramid of Human Needs

The purpose of design is ultimately about what it does for the user in the context of a work practice. It includes aid, support, capabilities, service, and even the joy the system provides the user. In this section, we provide a model for thinking about user needs, setting the stage to design for those needs.

There are at least three categories of human needs that must be met by a design:

- Ecological: Human users need to be able to participate in, and thrive in, the ecology of the work (or play) domain.
- Interaction: Human users need to be able to perform required tasks in the ecology of the domain.
- Emotional: Human users need to be emotionally and culturally satisfied and enriched as they use the product, including the need to form long-term emotional relationships with the product, which we call meaningfulness.

Designers often have to think about these categories in that order because they build on each other, as shown in the pyramid of human needs in the context of UX design (Fig. 11.5). When designers work in one of these layers, they operate in the perspective of that layer: the ecological perspective, the interaction perspective, and the emotional perspective.

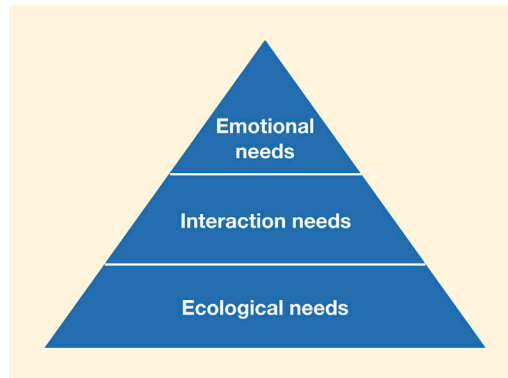


Fig. 11.5
Pyramid of human needs
that are the purpose of UX
design.

Meeting ecological needs is a prerequisite for meeting any other type of needs and is the foundational layer of the pyramid. Similarly, a design cannot satisfy users' emotional needs without first meeting ecological and interaction needs.

We will talk about designing the ecology (designing for a network or system of devices, physical locations, users, and information across which interactions can span) in [Chapter 15](#), designing the interaction in [Chapter 16](#), and designing for emotional impact in [Chapter 17](#).

11.3 STAGES OF DESIGN

Design is perhaps the most important lifecycle activity and the one with the broadest purview. Typically, the things you do in the design solutions lifecycle activity change dramatically over time as the project and the product evolve and mature through these basic stages (Fig. 11.6):

- Generative design (see [Chapter 13](#)): Ideation, sketching (very low-fidelity prototyping), and critiquing are used to generate new design ideas.
- Conceptual design (see [Chapter 14](#)): Mental models, system models, storyboards, low-fidelity prototypes are constructed for conceptual design candidates.

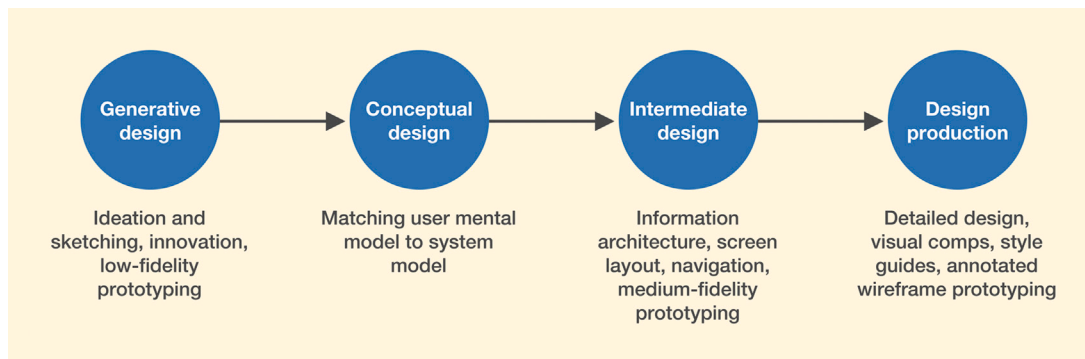


Fig. 11.6

Typical things you do in the design solutions lifecycle activity.

- Intermediate design: Ecological (see [Chapter 15](#)), interaction (see [Chapter 16](#)), and emotional designs (see [Chapter 17](#)) are developed for the most promising candidates, using illustrated scenarios, wireframes, medium-fidelity mockups of design forerunners, and identifying design tradeoffs to compare design candidates.
- Design production: Detailed design plans for implementation of the emerging design choice are specified.

11.4 ITERATION IN THE STAGES OF DESIGN

As a practical matter, the design solutions box in the overall UX lifecycle (see [Fig. 11.1](#)) unfolds as a series of iterative sub-lifecycles or activities ([Fig. 11.7](#)).

The observant reader will note that the progressive series of iterative loops in [Fig. 11.7](#) can be thought of as a spiral lifecycle concept (Boehm, 1988). Each loop, in turn, addresses an increasing level of detail and fidelity.

11.4.1 Deciding on the Design Goal

The focus of this step is to agree on the goal for all subsequent design iterations. The question to answer is this: Is the goal of design to create a solution that will support existing work practice, or is it to create a solution that will radically transform the work practice (see [Chapter 12](#))?

11.4.2 Generative Design Iteration

The focus of this phase of design creation is to generate as many design ideas and proposals as possible (see [Chapter 13](#)). Iteration to explore these design

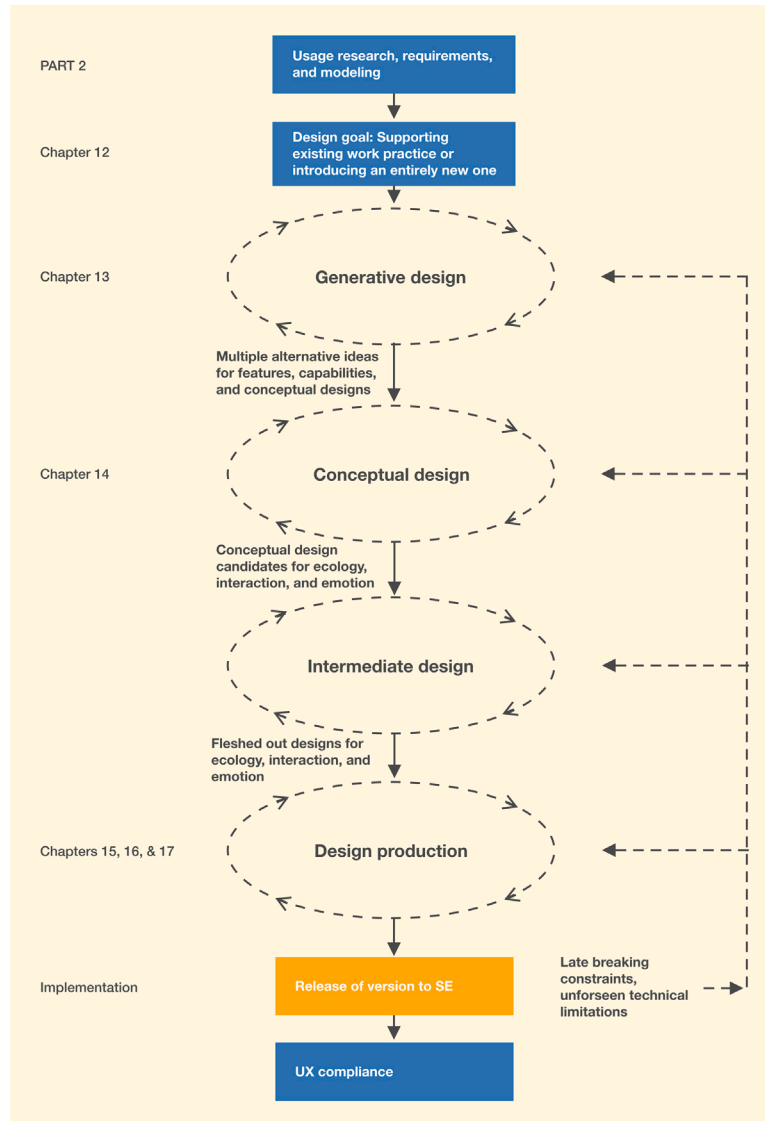


Fig. 11.7
Macro view of lifecycle activities
in design.

possibilities (Fig. 11.8) is a fast, loosely structured activity. This activity, a micro lifecycle within itself, includes synthesis, ideation, sketching, and critiquing.

The role of prototype is played by sketches, and the role of evaluation is carried out by discussion and critiquing. The output of generative design is a set of alternatives for conceptual designs and other capabilities or patterns for each level of the needs pyramid, mostly in the form of annotated rough sketches or storyboards.

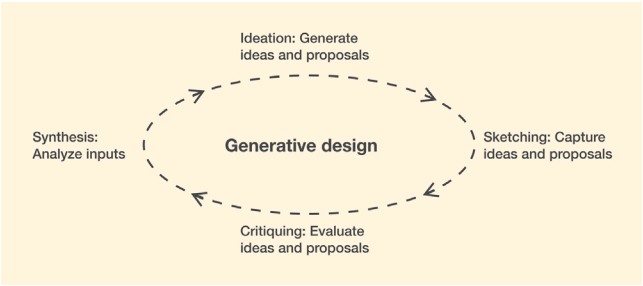


Fig. 11.8
Generative design iteration.

11.4.3 Conceptual Design Iteration

This iteration includes fleshing out the details of the high-level design theme or metaphor for each layer of the needs pyramid that came out of the generative design phase (Fig. 11.9).

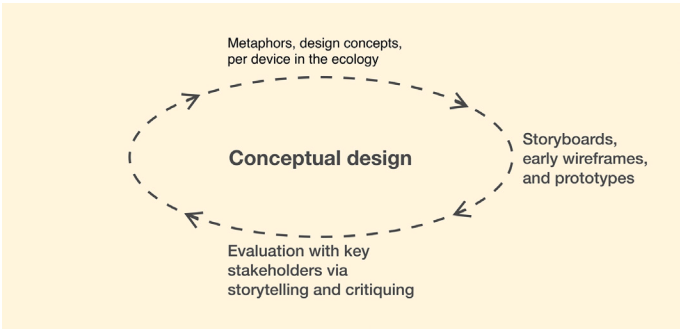


Fig. 11.9
Conceptual design iteration.

11.4.4 Intermediate Design Iteration

The purpose of intermediate design (Fig. 11.10) is to sort out possible multiple conceptual design candidates, to arrive at the one most viable for ecological, interaction, and emotional designs, and then to proceed to flesh out the details of each candidate.

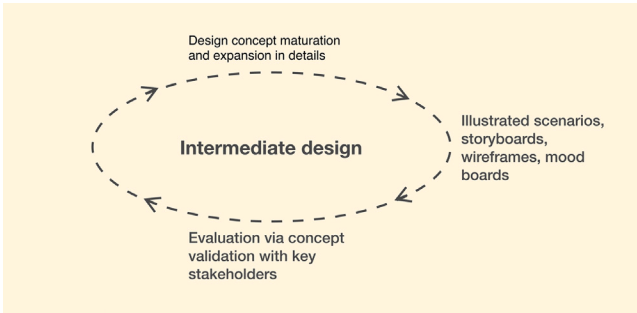


Fig. 11.10
Intermediate design iteration.

For example, for the Ticket Kiosk System, at least two conceptual design candidates for the kiosk's interaction design were explored further. One is a traditional navigational concept where users are shown available categories (e.g., movies, concerts, MU athletics) from which they drill down. Based on their choice on the first screen, users are shown further options and details to navigate. A second conceptual design is the one using the three-panel idea described in [Fig. 16.2](#).

Intermediate design, the activity that usually takes the longest, is where you work out the details of the envisioned ecology, create logical flows of intermediate-level navigational structure and screen designs for interaction, and flesh out themes for emotional impact.

During this phase, all layout and navigation elements become fully developed. The role of prototypes is played by wireflows (sequences of wireframes showing key workflows), illustrated scenarios, click-through mockups, and mood boards (collages of artifacts and images showcasing a theme of emotional impact to be embraced within a UX design). Using wireflows, or sequences of wireframes (see [Section 18.3](#)), key workflows are represented while describing what happens when the user interacts with the various user interface objects in the design. It is not uncommon to have wireframe sets represent part of the workflow or each task sequence using click-through prototypes.

Similarly, emotional impact issues, such as visual design styles, iconography, tone, typography, animation frameworks, and auditory styles, are all fleshed out. Much of this can fall to people with specialized training, the type of which you get in a design school or industrial design department.

One of the best ways to describe parts of your intermediate interaction design in a document is through illustrated scenarios, which combine the visual communication capability of storyboards and screen sketches with the capability of textual scenarios to communicate details. The result is an excellent vehicle for sharing and communicating designs to the rest of the team, and to management, marketing, and all other stakeholders.

Making illustrated scenarios is simple: Intersperse graphical storyboard frames and/or screen sketches as figures in the appropriate places to illustrate the narrative text of a design scenario. The storyboards in initial illustrated scenarios can be sketches or early wireframes.

The best way to communicate intermediate designs for emotional impact is through mood boards (see [Section 17.4.2](#)), visuals of sample screens, typographic palettes, and sound libraries.

As part of this stage, the designs are presented to cross-functional stakeholders (product managers and software engineering [SE] developers)

to get feedback about feasibility, platform constraints, etc., and for ensuring the emerging product scope in the designs is appropriate for the upcoming release version.

11.4.5 Detailed Design Iteration

Detailed design iteration (Fig. 11.11), often called design production, is where you iterate on the details of the design and finalize screen and layout details, including visual comps (pixel-perfect mockups of the graphical skin, including objects, colors, sizes, shapes, fonts, spacing, and location), for the look and feel of appearance for each screen on each device type in the ecology. This is also where you make any necessary changes before an official handoff to SE roles in the next phase.

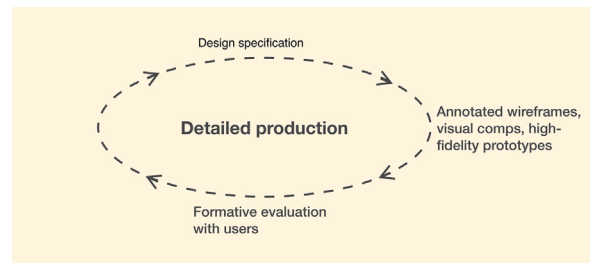


Fig. 11.11
Detailed design iteration
(design production).

The prototypes at this stage are usually detailed and annotated wireframes and/or high-fidelity interactive mockups. They include all user interface objects and data elements, represented with higher fidelity and annotated with callout text.

11.4.6 Software Engineering Implementation

In this phase, the SE counterparts begin implementation. In practice, late-breaking constraints or unforeseen technical limitations may be discovered by the SE role that kick the design specifications back to the UX team. Depending on the severity of the issue, UX designers may need to make changes and go back to an earlier iteration stage. For deep system-level issues, the UX team may have to go as far back as revisiting the conceptual design decisions to make adjustments. The most common changes at this stage, however, tend to be at the detailed design level and require small adjustments.

11.4.7 UX Fidelity Verification Phase

After the SE role completes the implementation, the UX role checks the final implementation to ensure that SE faithfully implemented the UX design they were given in the handoff.

11.5 DESIGN LIFECYCLE FOR THE AGILE UX FUNNEL

How does this design-creation sub-lifecycle fit into the agile UX funnel concept we discussed in [Chapter 4](#)? In [Fig. 11.12](#), we show another variation in the design sub-lifecycle: iteration in each slice of the agile UX funnel.

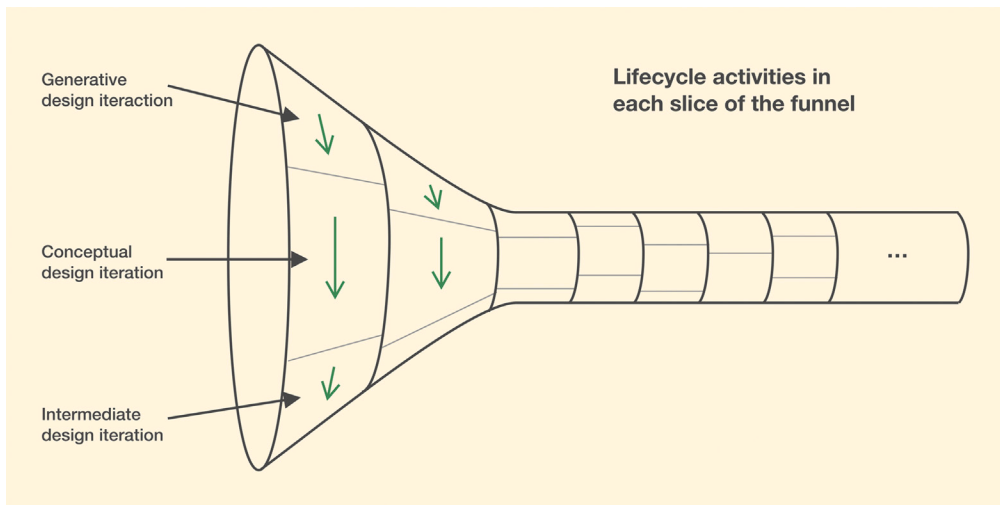


Fig. 11.12
Design-creation sub-lifecycle
in the agile UX funnel.

Designing each slice of functionality in the funnel entails an iteration of the steps in [Fig. 11.7](#). Work in the early funnel addresses a global view of the product or system and therefore has a large (systemwide) scope. This translates to more time spent on the generative design and conceptual design phases because the decisions of these iterations will have a major impact on the later slices of the funnel.

11.6 SEGUE

In this chapter, we discussed the nature of design. In the next chapter, we dive into the broader goals of a design effort.